INTERNSHIP PROJECT REPORT

(An Automated Reconnaissance Tool for DNS)

Organization

ITSOLERA Pvt. Ltd

Department

Offensive Security

Group/Team

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ABSTRACT

In modern penetration testing and bug bounty engagements, reconnaissance plays a critical role in identifying the attack surface of a target organization. Manual reconnaissance is often time-consuming, repetitive, and prone to human error. To address these challenges, this project presents ***Reconductor***, an automated reconnaissance tool designed to streamline and standardize the information-gathering phase of offensive security assessments.

Reconductor integrates multiple industry-standard reconnaissance tools into a single modular framework. It automates subdomain enumeration, DNS brute forcing, IP resolution, and technology stack identification while maintaining organized, timestamped outputs. This report documents the objectives, design, implementation, usage, and security relevance of the ***Reconductor*** tool developed as part of the internship at ***ITSOLERA Pvt. Ltd***.

1. Introduction

Reconnaissance is the first and one of the most crucial phases of penetration testing. The accuracy and depth of reconnaissance directly influence the success of later exploitation phases. Security professionals commonly rely on multiple open-source tools such as Amass, Subfinder, Assetfinder, and Wappalyzer to gather intelligence. However, using these tools independently leads to fragmented outputs and inefficiencies.

The ***Reconductor*** project was initiated to overcome these limitations by providing a unified, automated reconnaissance framework. The tool is specifically designed for penetration testers, red teamers, and bug bounty hunters to efficiently enumerate subdomains, resolve IP addresses, and identify technologies used by target domains.

2. Project Objectives

The primary objectives of the ***Reconductor*** project are:

- To automate the reconnaissance phase of penetration testing
- To integrate multiple passive and active subdomain enumeration tools
- To provide modular execution of reconnaissance components
- To generate well-structured, timestamped output directories
- To reduce manual effort and improve recon efficiency
- To ensure ethical and authorized usage for security testing

3. Project Scope

The scope of the ***Reconductor*** project is focused on automating and optimizing the reconnaissance phase of offensive security engagements. Reconnaissance is a foundational step in penetration testing, as it defines the visible attack surface of a target. ***Reconductor*** was designed to ensure this phase is comprehensive, repeatable, and efficient. The project scope includes the following key areas:

- Passive subdomain enumeration using open-source intelligence (OSINT) techniques
- Active subdomain discovery through DNS brute-force mechanisms
- Aggregation of results from multiple reconnaissance tools
- Deduplication and filtering of discovered subdomains
- Resolution of live subdomains to IP addresses
- Identification of web technologies and frameworks in use
- Structured storage of reconnaissance data for analysis and reporting

The project deliberately excludes vulnerability scanning, exploitation, privilege escalation, and post-exploitation activities. This limitation ensures that Reconductor remains a focused reconnaissance tool and complies with ethical and legal boundaries of authorized security testing.

4. System Requirements

4.1. Hardware Requirements

The tool is lightweight and can be executed on standard penetration testing environments. The minimum hardware requirements are:

- **RAM:** 4 GB Minimum (8 GB recommended for large scans)
- **Processor:** Dual-core CPU or Higher
- **Storage:** Minimum 10 GB free disk space for tools and scan outputs

4.2. Software Requirements

The following software components are required for successful execution:

- **Operating System:** Linux-based systems (Kali Linux preferred) or WSL
- **Language:** Python Version 3.8 or Higher
- **Git:** Version Control System required for managing repositories.
- **Docker (Optional):** For containerized execution

5. Tools & Technologies

Reconductor integrates several industry standard reconnaissance tools widely used by penetration testers and bug bounty hunters. Each tool contributes a specific capability to the overall reconnaissance workflow.

- **Python3:** Primary programming language used for orchestration and automation.
- **OWASP Amass:** Performs subdomain enumeration using multiple data sources.
- **Subfinder:** Provides fast and passive subdomain discovery
- **Assetfinder:** Identifies related domains and assets
- **crt.sh:** Extracts subdomains from SSL/TLS certificate transparency logs
- **dnscan:** Performs DNS brute-force enumeration using wordlists
- **dnsx:** Resolves discovered subdomains to IP addresses
- **Wappalyzer:** Detects web technologies, frameworks, and services

- **Docker:** Ensures consistent execution environments
- **Git:** Version control and dependency management

6. Architecture & Design

Reconductor follows a modular and scalable architecture designed to support flexibility, maintainability, and ease of use. Each reconnaissance function is implemented as an independent module, allowing the tool to adapt to different testing requirements.

6.1. High-Level Architecture

At a high level, ***Reconductor*** consists of:

- A central orchestration script responsible for input handling and execution flow
- A unified output management system for Multiple independent recon modules.
- The orchestrator processes user-provided command-line arguments.
- It determines which modules to execute and how to manage their outputs.
- It also manages dependencies between modules.

6.2. Modular Design Approach

Each reconnaissance capability is encapsulated within its own module. This design provides the following advantages:

- Individual modules can be executed independently
- New tools can be added with minimal changes to existing code
- Improved debugging and maintenance
- Reusability of modules across different workflows

Each module follows a common structure defined by a base class, ensuring consistency in execution and output handling.

7. Execution Workflow

When the user enables the complete workflow, ***Reconductor*** executes modules in a predefined sequence to maximize coverage and accuracy:

- Passive subdomain enumeration using Amass
- Fast discovery using Subfinder utility
- Asset identification using Assetfinder
- Certificate transparency analysis via crt.sh

- DNS brute-force enumeration using dnscan
- Merging and deduplication of subdomains
- DNS resolution to identify live hosts
- Technology stack detection using Wappalyzer

This sequential execution ensures that data from multiple sources is consolidated effectively.

8. Implementation Details

8.1. Dir & Output Structure

Reconductor automatically creates timestamped directories for each scan. This approach prevents data overwriting and allows testers to track historical scans. Each directory contains tool-specific outputs and a configuration summary file.

8.2. Command-Line Interface

The tool provides a flexible command-line interface that allows users to:

- Scan single or multiple target domains
- Enable or disable specific reconnaissance modules
- Supply custom wordlists for DNS brute-force
- Merge externally collected reconnaissance data

8.3. Output Files Generation

Key output files generated during a scan include:

- Tool-specific subdomain lists
- Consolidated and de-duplicated subdomain file
- Resolved IP address list
- Technology detection results in JSON format
- Scan configuration summary

9. Functional Workflow & Tool Execution

To clearly demonstrate the practical usability of ***Reconductor***, this section documents the functional workflow of the tool along with real execution evidence. Screenshots captured from the Kali Linux terminal during live runs can be placed in this section to visually validate each functionality.

9.1. Tool Initialization And Scan Configuration

This stage represents the initial execution of the ***Reconductor*** tool from the Kali Linux terminal. It includes providing command-line arguments such as the target domain, enabled modules, and optional parameters like custom wordlists.

At this step, ***Reconductor*** validates user input, prepares the execution environment, and creates a timestamped output directory. A scan configuration file is generated to log all parameters used during execution.

```
(DEMO) kali@LEOPARD-PC:~/Reconductor$ python3 reconductor.py -d google.com --crtsh --amass --subfinder --assetfinder --wappalyzer
################################################################################
    ____                                __                  __
   / __ \___  ________  ____  ____/ /_  _______/ /_____  _____
  / /_/ / _ \/ ___/ __ \/ __ \/ __  / / / / ___/ __/ __ \/ ___/
 / _, _/  __/ /__/ /_/ / / / / /_/ / /_/ / /__/ /_/ /_/ / /
/_/ |_|\___/\___/\____/_/ /_/\__,_/\__,_/\___/\__/\____/_/


[#>] An automated Python3 based DNS reconnaissance Framework - By TheLeopard65.
################################################################################
[+] Output directory: /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110
[✓] Scan configuration saved to /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/config.txt
[+] Running crt.sh enumeration
[+] Querying crt.sh for google.com
[+] crt.sh found 237 subdomains +---> /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/crtsh/crtsh.txt
[+] Running Amass passive enumeration
[-] Amass timed out for google.com
[+] Amass output saved to /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/amass/amass.txt
[+] Running Subfinder passive enumeration
[-] Subfinder timed out for google.com
[+] Subfinder output saved to /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/subfinder/subfinder.txt
[+] Running Assetfinder passive enumeration
[+] Assetfinder completed for google.com
[+] Assetfinder output saved to /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/assetfinder/assetfinder.txt
[!] Merging and filtering subdomains
[+] Merged 302 unique subdomains -> /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/subdomains.txt
[+] Detecting technologies with Wappalyzer
[+] about.google.com: Google Font API, Google Tag Manager
[+] actions.google.com: Google App Engine, Google Font API, Google Sign-in
[+] account.google.com: Google Font API, Google Tag Manager
[+] admob.google.com: AngularJS, Google Font API, Google Tag Manager
[+] 1.google.com: Google Font API
[+] adsense.google.com: Google Font API
[+] adscreativestudio.google.com: Google Analytics, Google Font API
[+] adstransparency.google.com: Google Font API
[+] ads.google.com: Bootstrap, Google Font API, Material Design Lite, jQuery, reCAPTCHA
[+] borgmon.evenflow-test.adz.google.com: Google Font API
[+] borgmon.evenflow.adz.google.com: Google Font API
[+] adwords.google.com: Bootstrap, Google Font API, Material Design Lite, jQuery, reCAPTCHA
[+] code.google.com: Google Font API, Material Design Lite
[+] chrome.google.com: Google Font API, Google Tag Manager
[+] corp.google.com: Google Font API
[+] google.com: Google Web Server
[+] services.google.com: Google Web Server
[+] developer.google.com: Google App Engine, Google Font API, Google Sign-in
[+] www.google.com: Google Web Server
[+] support.google.com: Google Analytics, Google Font API
[+] cloud.google.com: Google Font API
[+] m.google.com: Google Font API, Google Tag Manager
[+] www.adwords.google.com: Bootstrap, Google Font API, Material Design Lite, jQuery, reCAPTCHA
[+] www.ads.google.com: Bootstrap, Google Font API, Material Design Lite, jQuery, reCAPTCHA
[+] Wappalyzer results +---> /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110/wappalyzer/wappalyzer.json
(DEMO) kali@LEOPARD-PC:~/Reconductor$
```

9.2. Subdomain Enumeration & Discovery

During this phase, ***Reconductor*** executes multiple reconnaissance modules such as Amass, Subfinder, Assetfinder, crt.sh, and dnscan. Each tool runs sequentially, and their outputs are captured automatically.

This functionality highlights ***Reconductor's*** ability to aggregate results from multiple

enumeration techniques, improving coverage and accuracy compared to using individual tools manually.

9.3. Sub Domain Merging And IP Resolution

After enumeration, ***Reconductor*** merges all discovered subdomains into a single dataset, removes duplicates, and filters invalid entries. The filtered subdomains are then resolved to their corresponding IP addresses using dnsx. This step demonstrates how Reconductor converts raw reconnaissance data into actionable intelligence.

```
(DEMO) kali@LEOPARD-PC:~/Reconductor/output/scan_2026-04-12_14-19-31-031110$ cat subdomains.txt
*.apis.corp.google.com
*.appengine.google.com
*.auth.corp.google.com
*.bigstore-test.corp.google.com
*.bigstore.corp.google.com
*.blogger.corp.google.com
*.blogspot.corp.google.com
*.c.docs.google.com
*.c.pack.google.com
*.c.play.google.com
*.c.video.google.com
*.cache1.c.docs.google.com
*.cache1.c.play.google.com
*.cache1.c.video.google.com
*.cache2.c.docs.google.com
*.cache2.c.play.google.com
*.cache2.c.video.google.com
*.cache3.c.docs.google.com
```

```
(DEMO) kali@LEOPARD-PC:~/Reconductor/output/scan_2026-04-12_14-19-31-031110$ cat config.txt
##################################################
Scan Start time: 2026-04-12 14:19:31.034637
Target Domains: google.com
Specified Wordlist: /usr/share/seclists/Discovery/DNS/subdomains-top1million-20000.txt
Output Directory: /home/kali/Reconductor/output/scan_2026-04-12_14-19-31-031110
Subdomains File: None
DNS Resolution: Disabled
##################################################
Modules Enabled for Scan:
  - crtsh
  - amass
  - subfinder
  - assetfinder
  - wappalyzer
##################################################
(DEMO) kali@LEOPARD-PC:~/Reconductor/output/scan_2026-04-12_14-19-31-031110$
```

9.4. Technology Stack Identification

In the final functional stage, ***Reconductor*** uses Wappalyzer to identify technologies, frameworks, and services running on discovered web assets. The results are stored in structured *JSON* format for further analysis. This functionality assists security testers in understanding the target's technology stack, which is crucial for identifying potential attack vectors.

```
(DEMO) kali@LEOPARD-PC:~/Reconductor/output/scan_2026-04-12_14-19-31-031110$ cat wappalyzer/wappalyzer.json
{
  "about.google.com": [
    "Google Font API",
    "Google Tag Manager"
  ],
  "actions.google.com": [
    "Google App Engine",
    "Google Font API",
    "Google Sign-in"
  ],
  "account.google.com": [
    "Google Font API",
    "Google Tag Manager"
  ],
  "admob.google.com": [
    "AngularJS",
    "Google Font API",
    "Google Tag Manager"
  ],
  "1.google.com": [
    "Google Font API"
  ],
  "adsense.google.com": [
    "Google Font API"
  ],
  "adscreativestudio.google.com": [
    "Google Analytics",
    "Google Font API"
  ],
  "adstransparency.google.com": [
    "Google Font API"
  ],
  "ads.google.com": [
    "Bootstrap",
    "Google Font API",
    "Material Design Lite",
    "jQuery",
    "reCAPTCHA"
  ],
  "borgmon.evenflow-test.adz.google.com": [
    "Google Font API"
  ],
```

10. *Tool Usage Scenarios*

Reconductor can be effectively used in various offensive security scenarios:

- Initial attack surface mapping during penetration tests
- Bug bounty reconnaissance automation
- Security assessments for large organizations
- Training and educational demonstrations

11. *Ethical Considerations*

Ethical usage was a primary consideration during the development of ***Reconductor***. The tool is intended strictly for authorized testing.

- Only domains owned by the user or explicitly permitted should be scanned
- Rate limits and service policies must be respected

- No exploitation or intrusive actions are performed
- All activities must comply with legal and organizational policies

12. Results & Outcomes

The ***Reconductor*** tool successfully automates the reconnaissance phase by integrating multiple tools into a unified framework. The results demonstrate improved efficiency, consistency, and organization compared to manual reconnaissance workflows.

13. Limitations & De-merits

- Dependency on third-party tools and their updates
- Limited strictly to reconnaissance activities
- Requires proper environment configuration

14. Future Enhancements

Future improvements may include:

- Integration of vulnerability scanning modules
- Development of a web-based interface
- Cloud-based reconnaissance execution
- Advanced reporting and visualization features

15. Team Contribution

This project was successfully completed through the collective efforts of *Offensive Security (Team Alpha)* during the internship at ***ITSOLERA Pvt. Ltd***. Each team member contributed to different phases of the project, including research, tool integration, testing, documentation, and presentation preparation.

The collaborative environment and technical guidance provided by ***ITSOLERA Pvt. Ltd***. played a crucial role in the successful design and implementation of the ***Reconductor*** tool. The project reflects strong teamwork, practical offensive security skills, and adherence to professional and ethical standards.

16. Conclusion

Reconductor highlights the importance of automation in modern offensive security operations. By consolidating multiple reconnaissance tools into a modular and organized framework, the project delivers a practical solution for efficient attack surface discovery. The project also provided valuable real-world exposure to tool integration, automation, and professional security practices.

THE END